COTS Fine-Scale SDM: Comprehensive Validation Report

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Overview

This report validates the XGBoost COTS Fine-Scale Species Distribution Model using historical Manta Tow and Cull data. We compare both the **2025 Year-Specific** and **Year-Agnostic** models.

The validation involves: 1. **Grid-Based Historical Validation (100m)**: General statistics (ROC, AUC, Confusion Matrix) evaluating the model against historical COTS/scar presence. 2. **Tow-Path Simulation**: Operational simulation showing which manta tows would be suggested based on the model. 3. **Cull Dive Financial Savings**: Quantification of dive hours and dollars saved by avoiding ineffective culls (0.00 or <0.02 CPUE). 4. **Reef Maps**: Visual mapping of predictions vs. operations.

```r
library(terra)
library(sf)
library(dplyr)
library(readr)
library(readxl)
library(ggplot2)
library(tidyr)
library(knitr)
library(stringr)
library(pROC)

# --- Configuration ---
predict_agnostic_path <- "outputs/COTS_prob_0.02_cpue_agnostic.tif"
predict_year2025_path <- "outputs/COTS_prob_0.02_cpue_year2025.tif"
manta_csv <- "data/COTS Program  Manta Tow Data-2026-02-04.csv"
cull_xlsx <- "data/250929_COTS-Manta-Cull-RHIS-Data-Matthews-and-Schlawinsky.xlsx"

out_dir <- "validation_outputs"
dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)

# Selected Reefs
reef_select <- c(
  "Tongue Reef (16-026)",
  "Batt Reef (16-029)",
  "Arlington Reef (16-064)",
  "(Big) Broadhurst Reef (No 1) (18-100a)",
  "Britomart Reef (18-024)",
  "Sudbury Reef (No 1) (17-001a)",
  "Trunk Reef (18-027)",
  "Bramble Reef (18-029)",
  "Davies Reef (18-096)",
  "Heron Reef (23-052a)"
)
```

1. Load and Prepare Data

```r
# Load Model Predictions
prob_agnostic <- if(file.exists(predict_agnostic_path)) terra::rast(predict_agnostic_path) el
prob_year2025 <- if(file.exists(predict_year2025_path)) terra::rast(predict_year2025_path) el
```

```r
if(is.null(prob_year2025) || is.null(prob_agnostic)) stop("Missing prediction rasters.")

# Load Manta Tow Data
manta_raw <- read_csv(manta_csv, show_col_types = FALSE) %>%
  filter(
    !is.na(StartLongitude), !is.na(StartLatitude), !is.na(EndLongitude), !is.na(EndLatitude)
    StartLongitude != 0, StartLatitude != 0, EndLongitude != 0, EndLatitude != 0
  ) %>%
  mutate(
    cots_observed = CrownOfThornsStarfishCount > 0,
    scars_observed = !is.na(ScarsCount) & ScarsCount > 0,
    presence_any = cots_observed | scars_observed,
    ReefName = as.character(ReefName)
  )

# Convert Manta Tows to spatial lines
make_tow_lines <- function(df) {
  lines <- lapply(seq_len(nrow(df)), function(i) {
    p1 <- c(df$StartLongitude[i], df$StartLatitude[i])
    p2 <- c(df$EndLongitude[i], df$EndLatitude[i])
    st_linestring(rbind(p1, p2))
  })
  st_sf(df, geometry = st_sfc(lines, crs = 4326))
}

manta_sf <- make_tow_lines(manta_raw)
manta_sf <- st_transform(manta_sf, crs = crs(prob_year2025))
manta_selected <- manta_sf %>% filter(ReefName %in% reef_select)

# Load Cull Data
cull_raw <- read_excel(cull_xlsx, sheet = 4) %>%
  filter(!is.na(Longitude), !is.na(Latitude), !is.na(Bottomtime), Bottomtime > 0, Longitude
  filter(as.Date(SurveyDate) > as.Date("2018-11-30")) %>%
  mutate(
    Total = Cohort1 + Cohort2 + Cohort3 + Cohort4,
    CPUE = Total / Bottomtime,
    is_ghost = (Total == 0),
    is_non_problematic = (CPUE < 0.02),
    ReefName = as.character(ReefName)
  )
cull_selected <- cull_raw %>% filter(ReefName %in% reef_select)
cull_sf <- st_as_sf(cull_selected, coords = c("Longitude", "Latitude"), crs = 4326) %>%
```

```r
  st_transform(crs = crs(prob_year2025))

# Calculate Default Threshold: Median of 2025 raster cropped to selected reefs
ext_val <- terra::ext(manta_selected) + 2000
crop_2025 <- terra::crop(prob_year2025, ext_val)
```

```
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```

```r
median_thresh_2025 <- as.numeric(terra::global(crop_2025, fun=function(x) median(x, na.rm=TRU

crop_agnostic <- terra::crop(prob_agnostic, ext_val)
```

```
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```

```r
median_thresh_agnostic <- as.numeric(terra::global(crop_agnostic, fun=function(x) median(x, r

# Fixed mapping cutoffs
p_min_2025 <- 0.48
p_max_2025 <- 0.55

cat("Median Threshold (2025):", round(median_thresh_2025, 3), "\n")
```

```
Median Threshold (2025): 0.483
```

```r
cat("Median Threshold (Agnostic):", round(median_thresh_agnostic, 3), "\n")
```

```
Median Threshold (Agnostic): 0.562
```

```r
cat("Mapping Range (2025):", p_min_2025, "to", p_max_2025, "\n")
```

```
Mapping Range (2025): 0.48 to 0.55
```

2. Step A: Grid-Based Historical Validation (100m)

We rasterize historical manta tow observations onto a 100m grid and compare them to the SDM probability values (using both 2025 and Agnostic models). We calculate ROC curves, AUC, and evaluate thresholds using confusion matrices.

```
if(nrow(manta_selected) > 0) {
  # Aggregate SDM predictions to ~100m (factor of 3 -> 90m)
  pred_100m_2025 <- terra::aggregate(crop_2025, fact = 3, fun = "mean", na.rm = TRUE)
  pred_100m_agn <- terra::aggregate(crop_agnostic, fact = 3, fun = "mean", na.rm = TRUE)

  # Rasterize Manta Tow presence (1 if ever present, 0 if absent)
  manta_presence <- terra::rasterize(
    terra::vect(manta_selected),
    pred_100m_2025,
    field = "presence_any",
    fun = "max",
    background = NA
  )

  # Build evaluation dataframe
  val_df <- data.frame(
    observed_presence = terra::values(manta_presence)[,1],
    prob_2025 = terra::values(pred_100m_2025)[,1],
    prob_agnostic = terra::values(pred_100m_agn)[,1]
  ) %>% drop_na()

  # --- ROC and AUC ---
  roc_2025 <- roc(val_df$observed_presence, val_df$prob_2025, quiet=TRUE)
  roc_agn <- roc(val_df$observed_presence, val_df$prob_agnostic, quiet=TRUE)

  cat("AUC (2025 Model):", round(auc(roc_2025), 3), "\n")
  cat("AUC (Agnostic Model):", round(auc(roc_agn), 3), "\n\n")

  # Plot ROC
  par(mfrow=c(1,2))
  plot(roc_2025, main="ROC Curve (2025)", col="blue", lwd=2)
  plot(roc_agn, main="ROC Curve (Agnostic)", col="red", lwd=2)

  # --- Confusion Matrix at Median Threshold ---
  cat("Confusion Matrix for 2025 Model (Threshold = Median:", round(median_thresh_2025,3), ")
  cm_2025 <- table(
    Predicted = factor(val_df$prob_2025 >= median_thresh_2025, levels=c(FALSE, TRUE)),
```

```
    Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
  )
  print(cm_2025)
  cat("Accuracy:", sum(diag(cm_2025))/sum(cm_2025), "\n\n")

  cat("Confusion Matrix for Agnostic Model (Threshold = Median:", round(median_thresh_agnosti
  cm_agn <- table(
    Predicted = factor(val_df$prob_agnostic >= median_thresh_agnostic, levels=c(FALSE, TRUE))
    Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
  )
  print(cm_agn)
  cat("Accuracy:", sum(diag(cm_agn))/sum(cm_agn), "\n")
}
```

```
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AUC (2025 Model): 0.533
AUC (Agnostic Model): 0.62
```
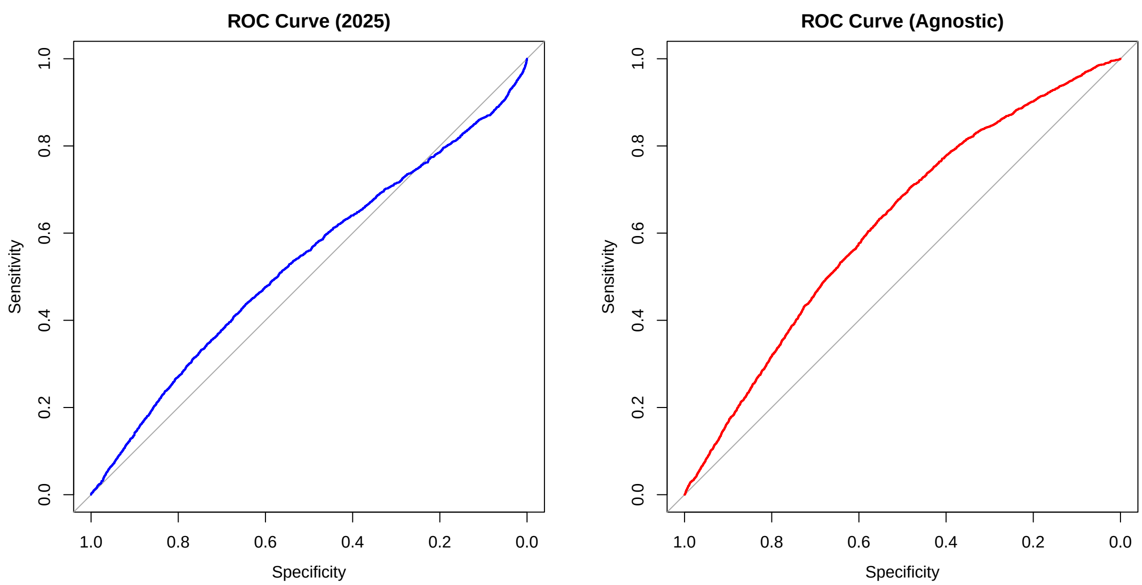



```
Confusion Matrix for 2025 Model (Threshold = Median: 0.483 )
         Observed
Predicted FALSE  TRUE
    FALSE 12696  2107
    TRUE   4590   745
Accuracy: 0.6674446

Confusion Matrix for Agnostic Model (Threshold = Median: 0.562 )
         Observed
Predicted FALSE  TRUE
    FALSE 11061  2307
    TRUE   6225   545
Accuracy: 0.5763234
```

3. Step B: Tow-Path Simulation and Evaluation

Simulating whether a specific tow path would have been suggested by the 2025 SDM model. We evaluate hit-rates strictly against tows conducted from Jan 1 2025 onwards, however, we use all historic tow tracks to delineate the operational boundaries.

```r
# 1. Calculate Global 2025 Raster Thresholds
thresh_2025_p88 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.88
thresh_2025_p95 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.95

cat("Global Thresholds for 2025 Validation:\n")
```

```
Global Thresholds for 2025 Validation:
```

```r
cat("- 88th Percentile:", round(thresh_2025_p88, 3), "\n")
```

```
- 88th Percentile: 0.506
```

```r
cat("- 95th Percentile:", round(thresh_2025_p95, 3), "\n\n")
```

```
- 95th Percentile: 0.515
```

```
# 2. Define Regions (100m Buffered Tows & 200m Grids)
tow_regions_list <- list()
grid_regions_list <- list()

for (r_name in unique(manta_selected$ReefName)) {
  m_sub <- manta_selected %>% filter(ReefName == r_name)
  if (nrow(m_sub) == 0) next

  bb <- st_bbox(m_sub)
  gr_500 <- st_make_grid(bb, cellsize = 500, square = TRUE) %>% st_as_sf()

  # Tow Regions (100m buffer, unioned, then intersected with 500m grid)
  tr <- st_buffer(m_sub, dist = 100) %>%
    st_union() %>%
    st_intersection(gr_500) %>%
    st_cast("POLYGON") %>%
    st_as_sf() %>%
    mutate(ReefName = r_name, RegionType = "Tow_100m_Seg500m", RegionID = paste0(r_name, "_T
    rename(geometry = x)
  tow_regions_list[[r_name]] <- tr

  # Grid Regions (200m x 200m)
  gr <- st_make_grid(bb, cellsize = 200, square = TRUE) %>%
    st_as_sf() %>%
    mutate(ReefName = r_name, RegionType = "Grid_200m", RegionID = paste0(r_name, "_G_", row
    rename(geometry = x)

  # Keep only grid cells that intersect with manta tows to avoid evaluating empty ocean
  gr <- gr[lengths(st_intersects(gr, m_sub)) > 0, ]

  grid_regions_list[[r_name]] <- gr
}

tow_regions <- bind_rows(tow_regions_list)
grid_regions <- bind_rows(grid_regions_list)
all_regions <- bind_rows(tow_regions, grid_regions)

# 3. Extract SDM Stats for Regions
extract_poly_stats <- function(r, polys) {
  v <- terra::vect(polys)
  ex <- terra::extract(r, v)
  val_col <- setdiff(names(ex), "ID")[1]
```

```
  ex %>%
    rename(poly_id = ID, value = !!val_col) %>%
    group_by(poly_id) %>%
    summarise(
      reg_p88    = as.numeric(quantile(value, probs = 0.88, na.rm = TRUE, names = FALSE)),
      reg_p95    = as.numeric(quantile(value, probs = 0.95, na.rm = TRUE, names = FALSE)),
      .groups    = "drop"
    )
}

all_regions$poly_id <- seq_len(nrow(all_regions))
region_stats <- extract_poly_stats(prob_year2025, all_regions)

all_regions <- all_regions %>% left_join(region_stats, by = "poly_id")
tow_regions <- all_regions %>% filter(RegionType == "Tow_100m_Seg500m")
grid_regions <- all_regions %>% filter(RegionType == "Grid_200m")

# 4. Map Tows to Regions and Apply Thresholds
# using st_join(largest = TRUE) ensures a tow is assigned to only one region
m_tow <- st_join(manta_selected, tow_regions %>% select(RegionID_Tow = RegionID, reg_p88_Tow
m_grid <- st_join(manta_selected, grid_regions %>% select(RegionID_Grid = RegionID, reg_p88_(

manta_eval <- manta_selected %>%
  mutate(
    reg_p88_Tow = m_tow$reg_p88_Tow,
    reg_p95_Tow = m_tow$reg_p95_Tow,
    reg_p88_Grid = m_grid$reg_p88_Grid,
    reg_p95_Grid = m_grid$reg_p95_Grid
  ) %>%
  mutate(
    sug_Tow_P88 = !is.na(reg_p88_Tow) & reg_p88_Tow >= thresh_2025_p88,
    sug_Tow_P95 = !is.na(reg_p95_Tow) & reg_p95_Tow >= thresh_2025_p95,
    sug_Grid_P88 = !is.na(reg_p88_Grid) & reg_p88_Grid >= thresh_2025_p88,
    sug_Grid_P95 = !is.na(reg_p95_Grid) & reg_p95_Grid >= thresh_2025_p95
  )

# Function to compute hit rates by reef for a specific scenario
calc_hit_rates_reef <- function(df, scenario_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
```

```
      Total_Tows = n(),
      Tows_Suggested = sum(!!sym(scenario_col), na.rm=TRUE),
      Hit_Rate = sum(!!sym(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
      Tows_Avoided = sum(! (!!sym(scenario_col)), na.rm=TRUE),
      Miss_Rate = sum(! (!!sym(scenario_col)) & presence_any, na.rm=TRUE) / Tows_Avoided
    )
  by_reef <- df %>%
    group_by(ReefName) %>%
    summarise(
      Scenario = scenario_name,
      Total_Tows = n(),
      Tows_Suggested = sum(!!sym(scenario_col), na.rm=TRUE),
      Hit_Rate = sum(!!sym(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
      Tows_Avoided = sum(! (!!sym(scenario_col)), na.rm=TRUE),
      Miss_Rate = sum(! (!!sym(scenario_col)) & presence_any, na.rm=TRUE) / Tows_Avoided
    )
  bind_rows(by_reef, overall)
}

manta_eval_2025 <- manta_eval %>% filter(as.Date(SurveyTime) >= as.Date("2025-01-01"))

hit_table <- bind_rows(
  calc_hit_rates_reef(manta_eval_2025, "sug_Tow_P88", "Seg Tow 100m + P88"),
  calc_hit_rates_reef(manta_eval_2025, "sug_Tow_P95", "Seg Tow 100m + P95")
) %>% arrange(ReefName, desc(Scenario))

kable(hit_table, digits = 3, caption = "Manta Tow Simulation: Operational Zonal Validation")
```

Table 1: Manta Tow Simulation: Operational Zonal Validation

ReefName	Scenario	Total_Tows	Tows_Suggested	Hit_Rate	Tows_Avoided	Miss_Rate	geometry
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P95	637	332	0.099	305	0.128	MULTILINESTRING ((573577.7 ...
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P88	637	185	0.076	452	0.128	MULTILINESTRING ((573577.7 ...
Arlington Reef (16-064)	Seg Tow 100m + P95	328	145	0.193	183	0.022	MULTILINESTRING ((406118.2 ...

ReefName	Scenario	Total_Tows	Tows_Suggested	Hit_Rate	Tows_Avoided	Miss_Rate	geometry
Arlington Reef (16-064)	Seg Tow 100m + P88	328	108	0.250	220	0.023	MULTILINESTRING ((406118.2 …
Batt Reef (16-029)	Seg Tow 100m + P95	382	344	0.308	38	0.026	MULTILINESTRING ((377707.6 …
Batt Reef (16-029)	Seg Tow 100m + P88	382	317	0.312	65	0.123	MULTILINESTRING ((377707.6 …
Britomart Reef (18-024)	Seg Tow 100m + P95	162	76	0.000	86	0.000	MULTILINESTRING ((467946.8 …
Britomart Reef (18-024)	Seg Tow 100m + P88	162	43	0.000	119	0.000	MULTILINESTRING ((467946.8 …
Davies Reef (18-096)	Seg Tow 100m + P95	177	100	0.200	77	0.260	MULTILINESTRING ((566203.3 …
Davies Reef (18-096)	Seg Tow 100m + P88	177	60	0.117	117	0.282	MULTILINESTRING ((566203.3 …
Heron Reef (23-052a)	Seg Tow 100m + P95	217	163	0.147	54	0.093	MULTILINESTRING ((1003623 7…
Heron Reef (23-052a)	Seg Tow 100m + P88	217	154	0.149	63	0.095	MULTILINESTRING ((1003623 7…
Overall	Seg Tow 100m + P95	2619	1491	0.161	1128	0.075	MULTILINESTRING ((376792.6 …
Overall	Seg Tow 100m + P88	2619	1109	0.174	1510	0.087	MULTILINESTRING ((376792.6 …
Tongue Reef (16-026)	Seg Tow 100m + P95	716	331	0.088	385	0.042	MULTILINESTRING ((376792.6 …
Tongue Reef (16-026)	Seg Tow 100m + P88	716	242	0.095	474	0.046	MULTILINESTRING ((376792.6 …

4. Step C: Cull Dive Financial Savings

Evaluating cull dives strictly from Jan 1 2025 onwards using the 2025 P88 and P95 thresholds on Segmented Tow Regions. We quantify avoided diving hours for dives that resulted in 0.00 CPUE (Ghost Culls) and <0.02 CPUE (Non-Problematic), and calculate financial savings ($950/hr).

```
# Map Cull Dives to Regions
c_tow <- st_join(cull_sf, tow_regions %>% select(RegionID_Tow = RegionID, reg_p88_Tow = reg_
c_grid <- st_join(cull_sf, grid_regions %>% select(RegionID_Grid = RegionID, reg_p88_Grid = 

cull_eval <- cull_sf %>%
  st_drop_geometry() %>%
  mutate(
    reg_p88_Tow = c_tow$reg_p88_Tow,
    reg_p95_Tow = c_tow$reg_p95_Tow,
    reg_p88_Grid = c_grid$reg_p88_Grid,
    reg_p95_Grid = c_grid$reg_p95_Grid
  ) %>%
  mutate(
    # Note: If a cull doesn't fall in any region, reg_p88 is NA, so it is "avoided" by defaul
    avoid_Tow_P88 = is.na(reg_p88_Tow) | reg_p88_Tow < thresh_2025_p88,
    avoid_Tow_P95 = is.na(reg_p95_Tow) | reg_p95_Tow < thresh_2025_p95,
    avoid_Grid_P88 = is.na(reg_p88_Grid) | reg_p88_Grid < thresh_2025_p88,
    avoid_Grid_P95 = is.na(reg_p95_Grid) | reg_p95_Grid < thresh_2025_p95
  )

diver_hour_cost <- 950

calc_savings_reef <- function(df, avoid_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
      Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_
      Total_Savings = Ghost_Savings + NonProb_Savings,
      Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
```

```
    ) %>%
    mutate(
      CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
    )
  by_reef <- df %>%
    group_by(ReefName) %>%
    summarise(
      Scenario = scenario_name,
      Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_h
      Total_Savings = Ghost_Savings + NonProb_Savings,
      Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
    ) %>%
    mutate(
      CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
    )
  bind_rows(by_reef, overall)
}

cull_eval_2025 <- cull_eval %>% filter(as.Date(SurveyDate) >= as.Date("2025-01-01")) %>%
  mutate(ManagementZone = case_when(
    ReefName %in% c("Tongue Reef (16-026)", "Batt Reef (16-029)", "Arlington Reef (16-064)",
    ReefName %in% c("Britomart Reef (18-024)", "Trunk Reef (18-027)", "Bramble Reef (18-029)"
    ReefName %in% c("Heron Reef (23-052a)") ~ "Southern",
    TRUE ~ "Far Northern"
  ))

savings_table <- bind_rows(
  calc_savings_reef(cull_eval_2025, "avoid_Tow_P88", "Seg Tow 100m + P88"),
  calc_savings_reef(cull_eval_2025, "avoid_Tow_P95", "Seg Tow 100m + P95")
) %>% arrange(ReefName, desc(Scenario))

kable(savings_table, digits = 2, format.args = list(big.mark = ","), caption = "Financial Sa
```

Table 2: Financial Savings and Risk Summary across Scenarios

ReefName	Scenario	Ghost_Avoided	Ghost_Savings	ProbNon_Avoided	ProbNon_Savings	Total_Savings	Prob_Missed	COTS_Missed	Pct_COTS_Missed	Hours_Missed	CPUE_Missed
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P95	16	62,684.17	37	166,044.17	228,728.33	14	185	16.59	83.33	0.04
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P88	23	87,764.17	50	211,770.83	299,535.00	15	189	16.95	86.33	0.04
Batt Reef (16-029)	Seg Tow 100m + P95	0	0.00	0	0.00	0.00	0	0	0.00	0.00	0.00
Batt Reef (16-029)	Seg Tow 100m + P88	1	2,533.33	2	4,908.33	7,441.67	15	303	6.88	80.00	0.06
Britomart Reef (18-024)	Seg Tow 100m + P95	0	0.00	0	0.00	0.00	0	0	0.00	0.00	0.00
Britomart Reef (18-024)	Seg Tow 100m + P88	0	0.00	2	14,250.00	14,250.00	2	21	36.21	15.00	0.02
Davies Reef (18-096)	Seg Tow 100m + P95	1	3,483.33	12	39,963.33	43,446.67	13	94	43.52	49.87	0.03
Davies Reef (18-096)	Seg Tow 100m + P88	1	3,483.33	12	39,963.33	43,446.67	13	94	43.52	49.87	0.03
Heron Reef (23-052a)	Seg Tow 100m + P95	1	5,510.00	5	22,277.50	27,787.50	1	11	3.26	5.33	0.03

ReefName	Scenario	Ghost_Avoided	Ghost_Savings	NonProb_Avoided	NonProb_Savings	Total_Savings	Prob_Missed	COTS_Missed	Pct_COTS_Missed	Hours_Missed	CPUE_Missed
Heron Reef (23-052a)	Seg Tow 100m + P88	1	5,510.00	6	26,647.50	32,157.50	1	11	3.26	5.33	0.03
Overall	Seg Tow 100m + P95	18	71,677.50	54	228,285.00	299,962.50	28	290	4.63	138.53	0.03
Overall	Seg Tow 100m + P88	26	99,290.83	72	297,540.00	396,830.83	36	618	9.86	236.53	0.04
Tongue Reef (16-026)	Seg Tow 100m + P95	0	0.00	0	0.00	0.00	0	0	0.00	0.00	0.00
Tongue Reef (16-026)	Seg Tow 100m + P88	0	0.00	0	0.00	0.00	0	0	0.00	0.00	0.00

```r
# Management Zone Summary
calc_savings_zone <- function(df, avoid_col, scenario_name) {
  df %>%
    group_by(ManagementZone) %>%
    summarise(
      Scenario = scenario_name,
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_
      Total_Savings = Ghost_Savings + NonProb_Savings,
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60,
      .groups = "drop"
    )
}

savings_zone_table <- bind_rows(
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P88", "Seg Tow 100m + P88"),
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P95", "Seg Tow 100m + P95")
```

```
) %>% arrange(ManagementZone, desc(Scenario))

kable(savings_zone_table, digits = 2, format.args = list(big.mark = ","), caption = "Financia
```

Table 3: Financial Savings and Risk Summary by Management Zone (2025)

ManagementZone	Scenario	Ghost_Savings	NonProb_Savings	Total_Savings	COTS_Missed	Pct_COTS_Missed	Hours_Missed
Central	Seg Tow 100m + P95	66,167.50	206,007.50	272,175.00	279	20.09	133.20
Central	Seg Tow 100m + P88	91,247.50	265,984.17	357,231.67	304	21.89	151.20
Northern	Seg Tow 100m + P95	0.00	0.00	0.00	0	0.00	0.00
Northern	Seg Tow 100m + P88	2,533.33	4,908.33	7,441.67	303	6.68	80.00
Southern	Seg Tow 100m + P95	5,510.00	22,277.50	27,787.50	11	3.26	5.33
Southern	Seg Tow 100m + P88	5,510.00	26,647.50	32,157.50	11	3.26	5.33

5. Step D: Visual Mapping

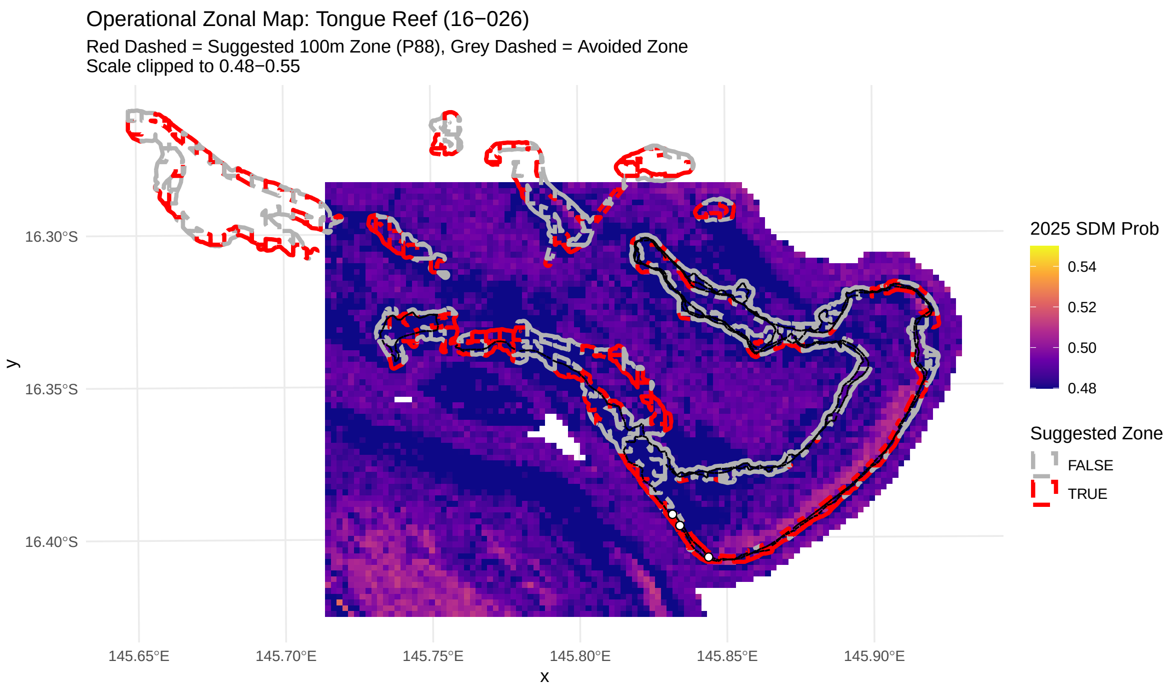
Below are maps for each reef showing the underlying 2025 SDM probability. The operational units displayed are the **100m Buffered Tow Regions**, categorized by the **88th Percentile (P88) Threshold** (Red = Suggested, Grey = Avoided). Individual manta tows and cull dive locations are also plotted to show what falls inside the suggested operational areas.

```
plot_reef_map <- function(r, tow_sf, dive_sf, region_sf, reef_name, p_min, p_max) {
  # Bounding box around features
  bb_t <- if(nrow(tow_sf) > 0) st_bbox(tow_sf) else NULL
  bb_d <- if(nrow(dive_sf) > 0) st_bbox(dive_sf) else NULL

  xmin <- min(c(bb_t["xmin"], bb_d["xmin"]), na.rm=TRUE)
  ymin <- min(c(bb_t["ymin"], bb_d["ymin"]), na.rm=TRUE)
```

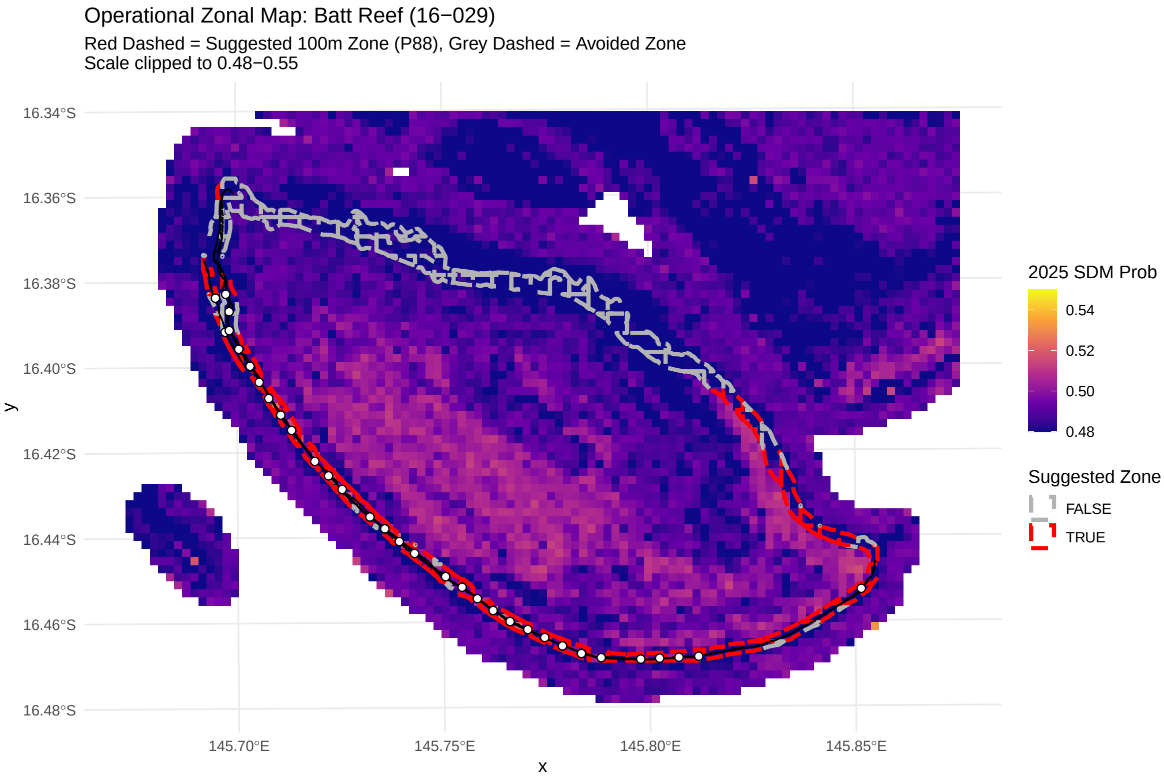
```
xmax <- max(c(bb_t["xmax"], bb_d["xmax"]), na.rm=TRUE)
ymax <- max(c(bb_t["ymax"], bb_d["ymax"]), na.rm=TRUE)

pad <- 2000 # 2km padding
ext_map <- terra::ext(xmin - pad, xmax + pad, ymin - pad, ymax + pad)

rc <- terra::crop(r, ext_map)

# Scale aggregate factor to keep cell count manageable (~10,000 pixels) for fast plotting
fact <- ceiling(sqrt(terra::ncell(rc) / 10000))
if(fact > 1) {
  rc <- terra::aggregate(rc, fact = fact, fun = "mean", na.rm = TRUE)
}

rdf <- as.data.frame(rc, xy = TRUE, na.rm = TRUE)
names(rdf)[3] <- "prob"

# Determine if region is suggested based on P88 threshold
region_sf <- region_sf %>%
  mutate(is_suggested = !is.na(reg_p88) & reg_p88 >= thresh_2025_p88)

gg <- ggplot() +
  geom_raster(data = rdf, aes(x = x, y = y, fill = prob)) +
  # Plot Operational Zones
  geom_sf(data = region_sf, aes(color = is_suggested), fill = NA, linewidth = 1.2, linetype
  scale_color_manual(values = c("TRUE" = "red", "FALSE" = "grey70")) +
  # Plot Tows and Dives
  geom_sf(data = tow_sf, color = "black", linewidth = 0.5) +
  geom_sf(data = dive_sf, shape = 21, fill = "white", color = "black", size = 2) +
  scale_fill_viridis_c(
    option = "C",
    na.value = NA,
    limits = c(p_min, p_max),
    oob = scales::squish
  ) +
  theme_minimal() +
  labs(
    title = paste("Operational Zonal Map:", reef_name),
    subtitle = paste0("Red Dashed = Suggested 100m Zone (P88), Grey Dashed = Avoided Zone\n
    fill  = "2025 SDM Prob",
    color = "Suggested Zone"
  )
```

```r
  return(gg)
}

# Generate plots for each reef
cull_sf_2025 <- cull_sf %>% filter(as.Date(SurveyDate) >= as.Date("2025-01-01"))

for(r_name in reef_select) {
  t_sf <- manta_eval_2025 %>% filter(ReefName == r_name)
  d_sf <- cull_sf_2025 %>% filter(ReefName == r_name)
  r_sf <- tow_regions %>% filter(ReefName == r_name)

  if(nrow(t_sf) > 0 || nrow(d_sf) > 0) {
    message(paste("Processing map for:", r_name))
    cat("\n### ", r_name, "\n")
    p <- plot_reef_map(prob_year2025, t_sf, d_sf, r_sf, r_name, p_min_2025, p_max_2025)
    print(p)
    cat("\n")
  }
}
```

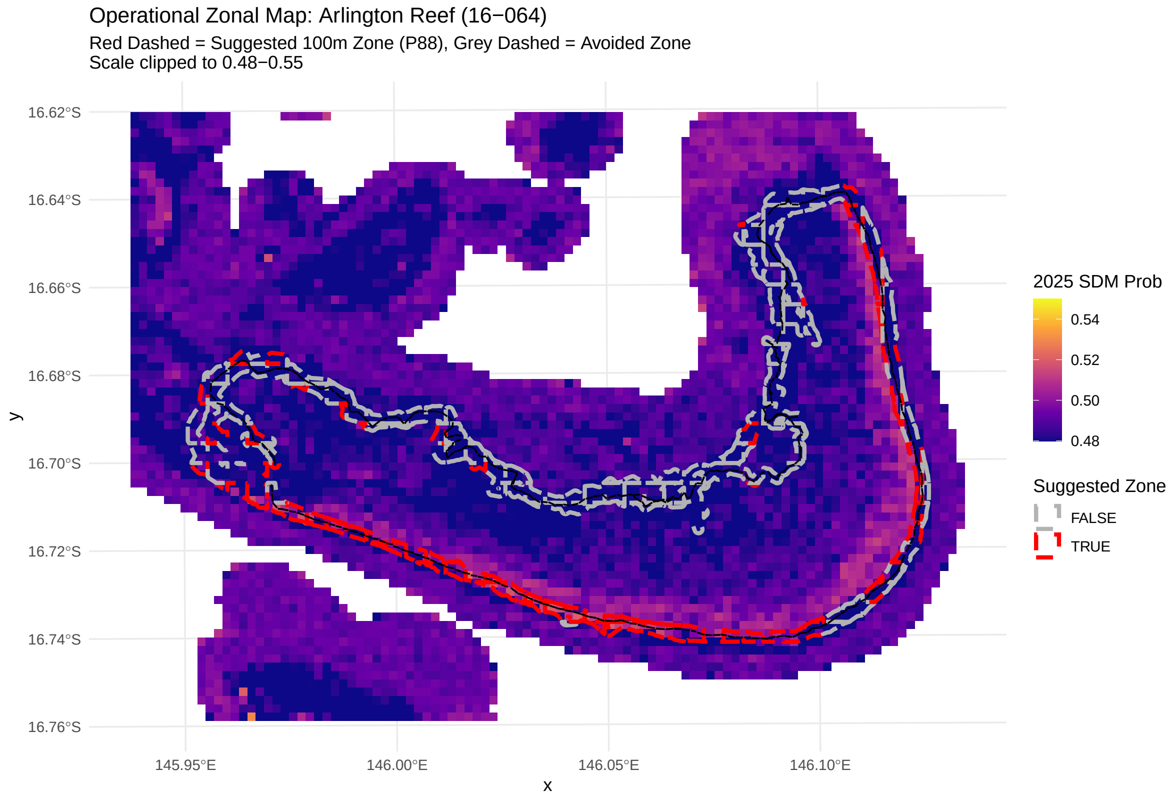
Tongue Reef (16-026)



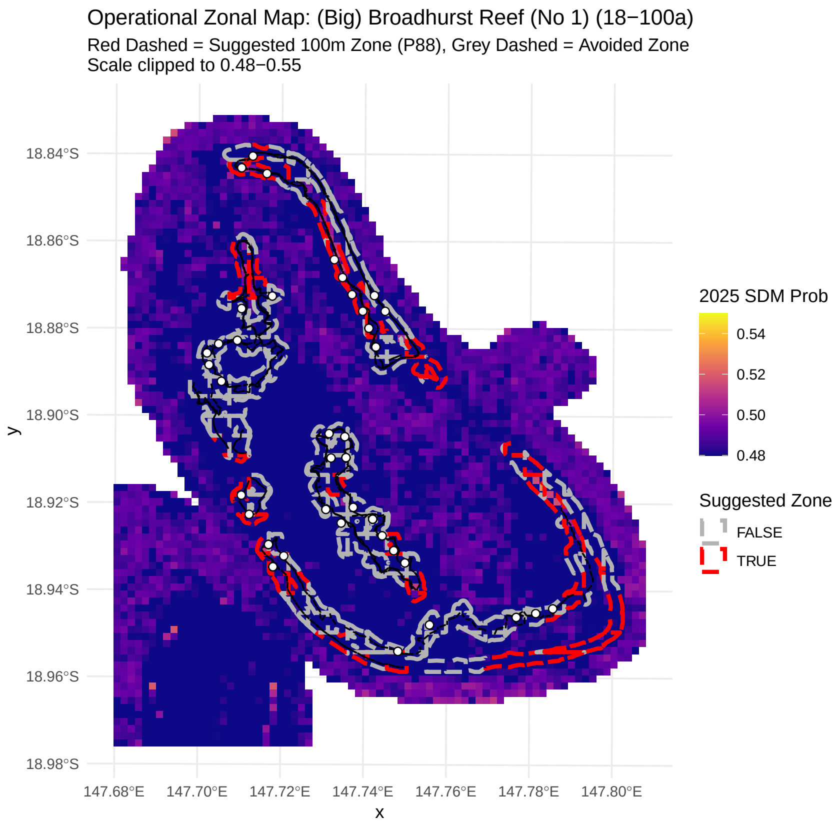
Batt Reef (16-029)



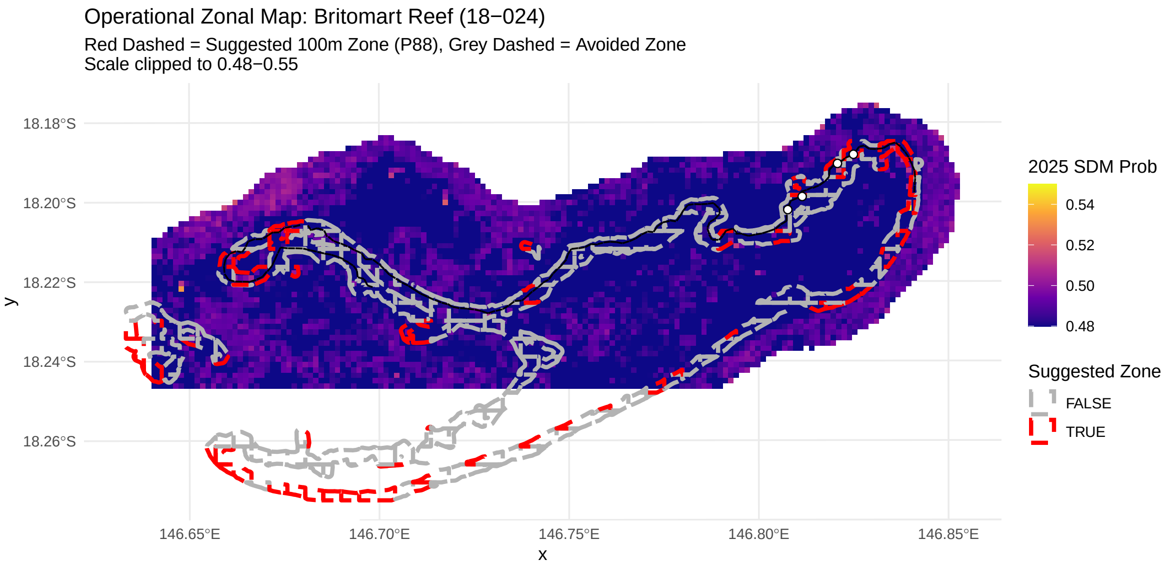
Arlington Reef (16-064)



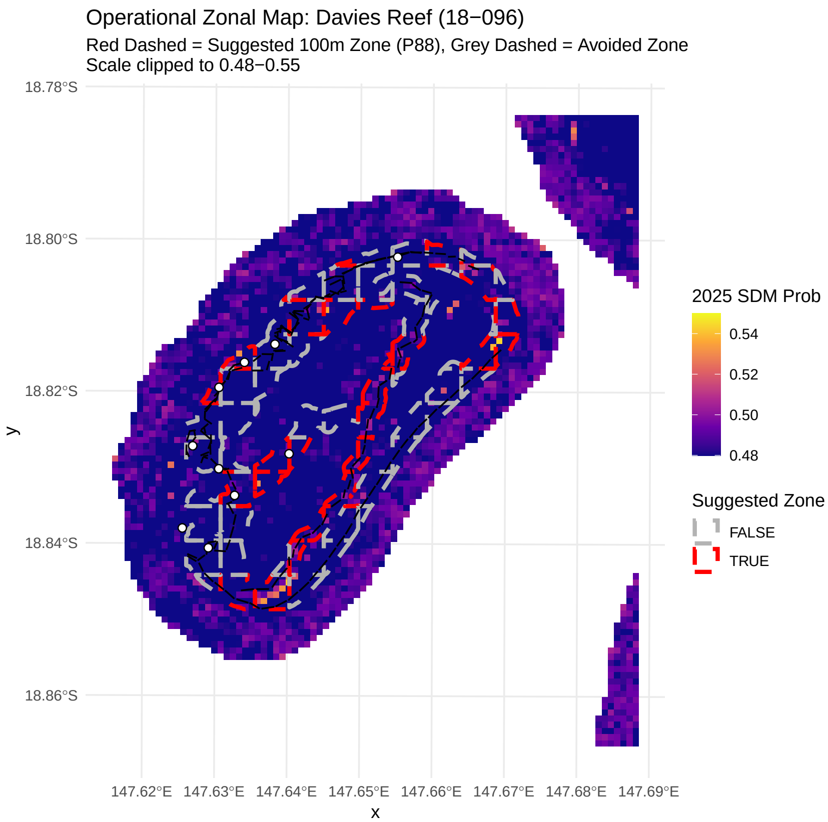
(Big) Broadhurst Reef (No 1) (18-100a)



Britomart Reef (18-024)



Davies Reef (18-096)



Heron Reef (23-052a)